

Face Recognition-Based Attendance System

Using Python

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1. Introduction

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Project Overview

- Briefly introduce the project.
- Explain the need for an automated attendance system.
- Mention the use of Face Recognition Technology to improve accuracy.



2. Problem Statement

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Issues with Traditional Systems

• Manual attendance systems are time-consuming and prone to fraud.
• Students can mark proxies in traditional methods.
• Need for a contactless, accurate, and efficient solution.



3. Objectives



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Goals of the System

- Automate attendance marking using facial recognition.
- Store attendance records securely in an Excel sheet.
- Allow real-time attendance marking for 7 periods per day.
- Prevent attendance marking during break (3:50-4:00 PM) and lunch (1:20-2:10 PM).

4. Technologies Used

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Technical Stack

- Python (Core language)
- OpenCV (For face detection)
- face_recognition (For face matching)
- Tkinter (For GUI interface)
- Pandas & OpenPyXL (For attendance record handling)





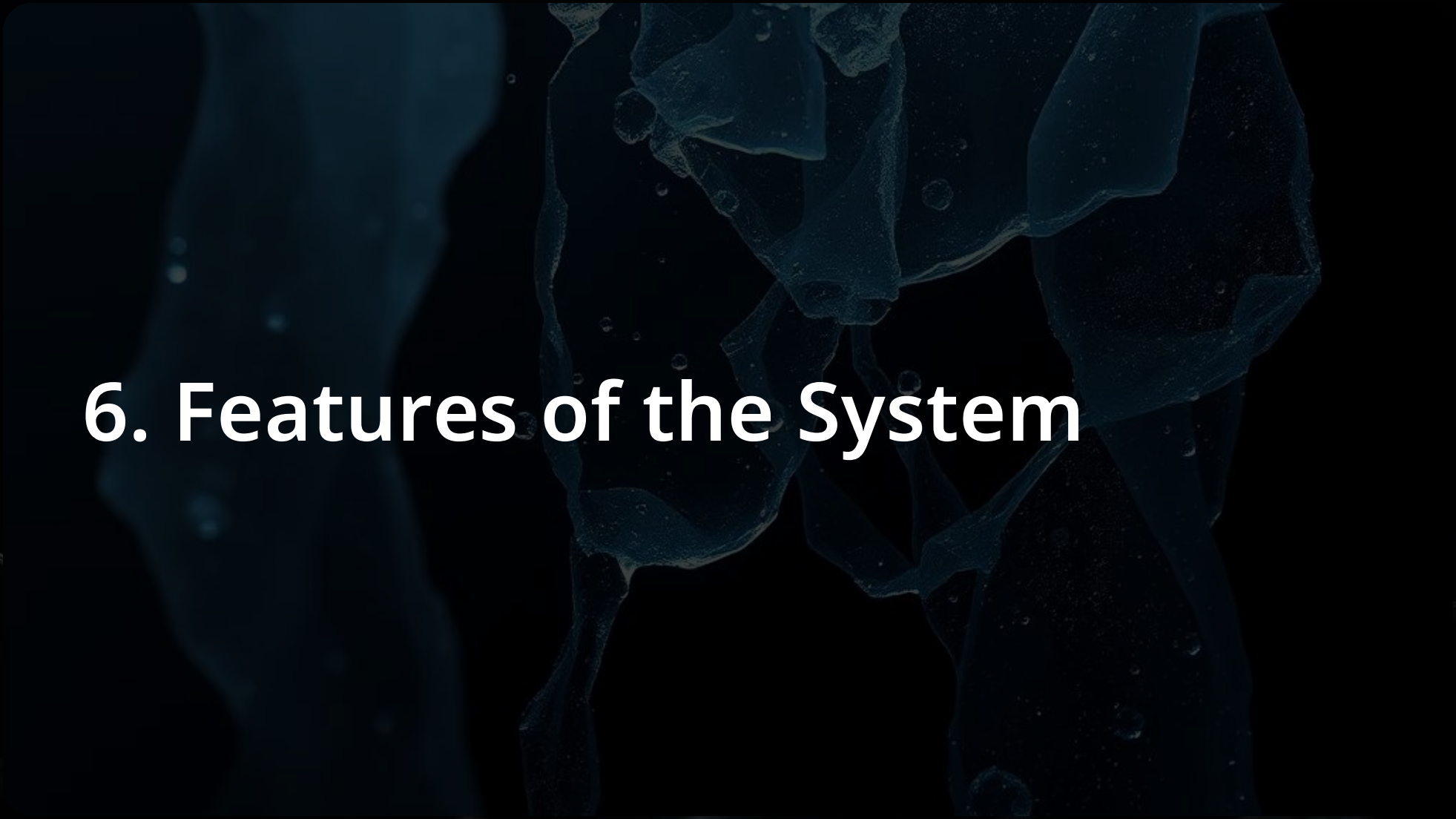
5. System Architecture



Process Flow

Steps Involved

- Step 1: Capture face images and store encodings.
- Step 2: Compare live faces with stored encodings.
- Step 3: If matched, mark attendance in attendance_YYYY-MM-DD.xlsx.
- Step 4: Attendance updates in the same row per period.
- Step 5: Allow users to check attendance records.



6. Features of the System



6. Features of the System

System Capabilities

- Face Recognition-Based – No manual entry required.
- Excel-Based Storage – Records can be downloaded.
- Multiple Period Tracking – Attendance marked for 7 periods daily.
- Missed Period Handling – Auto-fills absent periods with ❌.
- Duplicate Prevention – Attendance cannot be marked twice in a period.

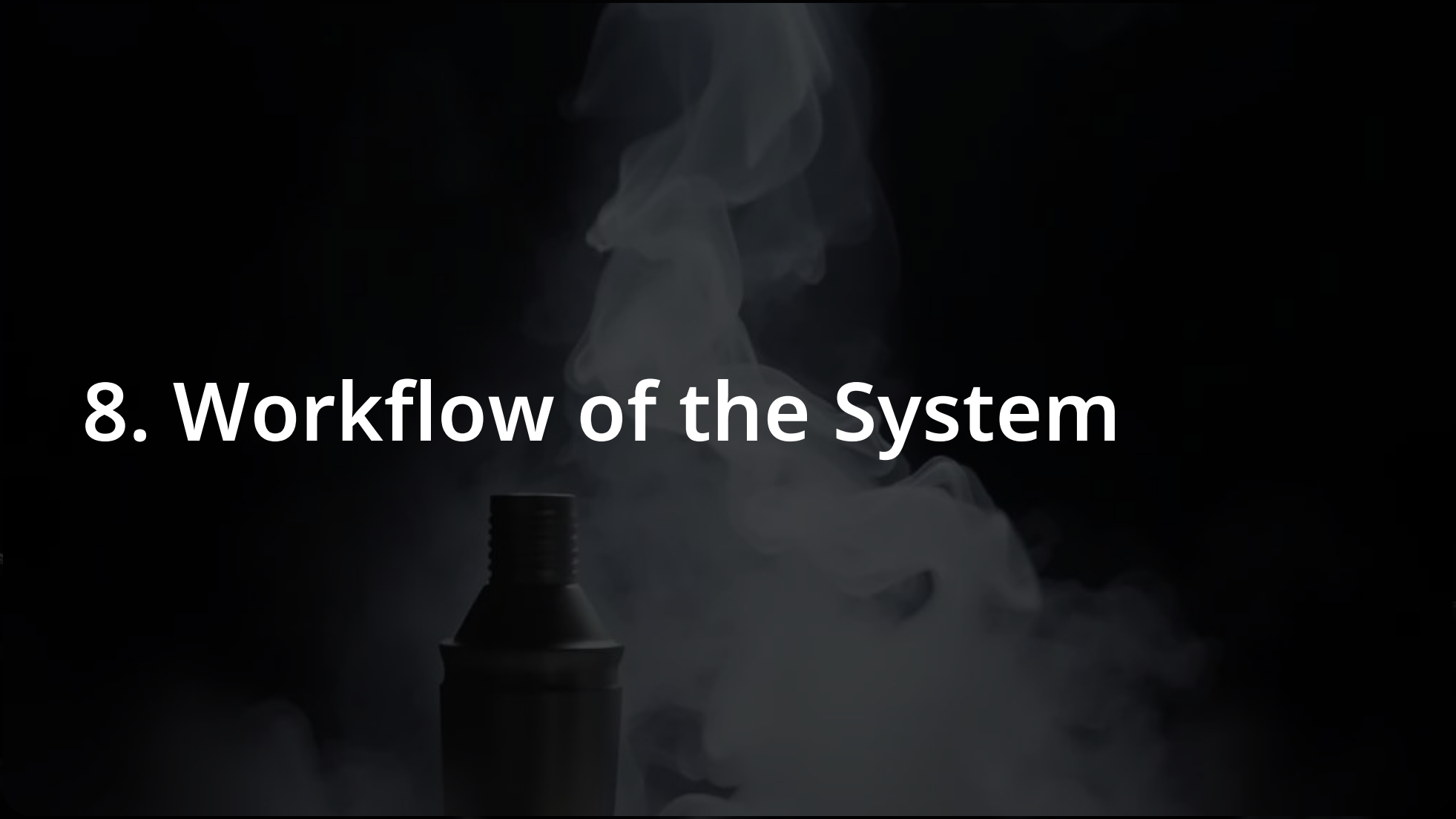


7. GUI Design

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User Interface

- Home page with two main options:
- Take Attendance – Starts face recognition.
- Add New Person – Allows adding a new student.
- Pop-up window for name & register number entry.



8. Workflow of the System

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Operational Steps

• Step 1: Capture student’s face via webcam.
• Step 2: Recognize face and fetch details from the database.
• Step 3: Check if the student is already marked for the current period.
• Step 4: If not marked, update attendance in the same row.
• Step 5: Auto-save attendance in an Excel file.

9. Output



9. Output

Visual Representation

- Live face recognition process.
- GUI interface with attendance marking.
- Attendance Excel file format.

10. Advantages of the System

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Benefits

- Eliminates proxy attendance.
- Increases accuracy & reduces errors.
- Saves time for teachers & students.
- Works in real-time.





11. Challenges & Solutions

Addressing Issues

Identified Challenges

- Challenge: Low light conditions.
- Solution: Enhanced image processing techniques.
- Challenge: Similar-looking students.
- Solution: High-accuracy facial encoding.



12. Future Enhancements

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Potential Improvements

- Cloud Storage Integration – Store attendance data online.
- Mobile App Interface – For remote attendance access.
- Multi-Camera Support – Cover multiple classrooms.





13. Conclusion



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Summary

• Summarize the importance of automation in attendance systems.
• Highlight the system’s efficiency & future scope.

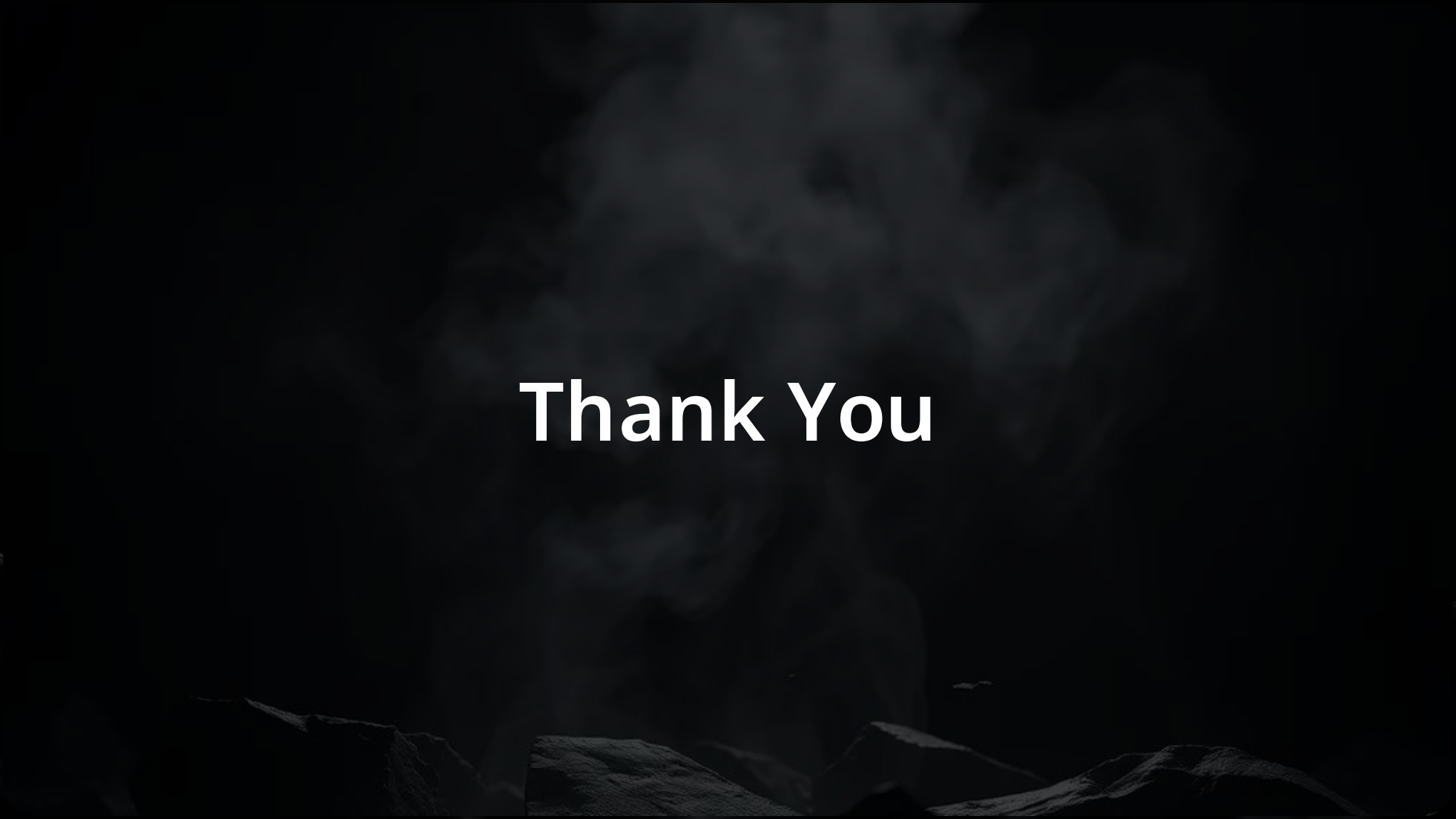

14. Q&A



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Discussion

- Open for discussion/questions.


Thank You